BUILDING ENERGY ANALYTICS

Heat Check

Find which buildings waste heat, and what to fix first.

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Heating is the biggest energy use, waste hides in it

32%

of US commercial building energy goes to space heating

Teams see how much a building uses, but not whether that is too much for the weather.

Heat left on at night or on warm days blends into a cold week.

With 100 buildings, nobody can inspect them all.

Source: US EIA, Commercial Buildings Energy Consumption Survey, 2018

Out of 97 buildings, it found the 10 worth a visit

10

buildings with confirmed waste, sent for human review

$159K

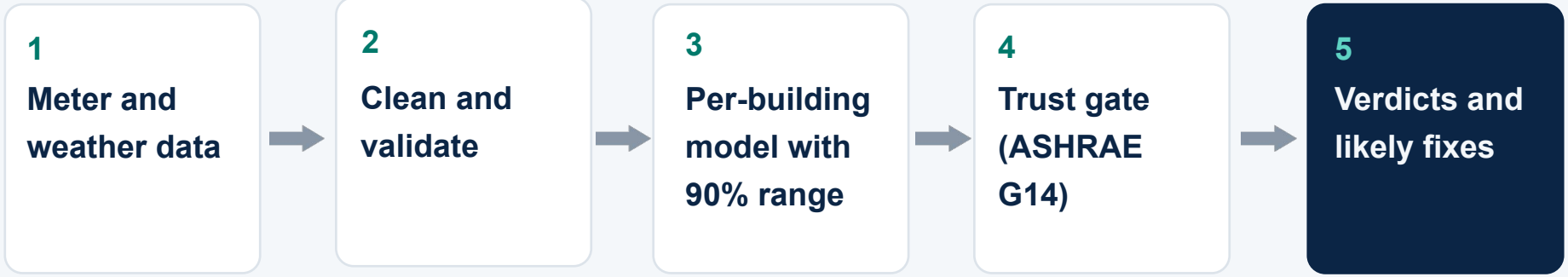
per year in wasted heating, measured conservatively

576 t

of CO2 per year, mostly fixable with low-cost controls changes

97 buildings, cold-climate campus, Building Data Genome 2 (Miller et al., 2020). Trained on 2016, results for 2017.

How it works



Python

Gradient-boosted trees

Conformal prediction

FastAPI

Docker

Render

CI with property-based tests

Built solo in Kiro in under eight hours

Why trust it, and the gap it fills

Built for trust

- Every estimate has a calibrated 90% range
- Models must pass the ASHRAE accuracy standard
- Says "Can’t judge" instead of guessing
- Property-based tests caught 2 real bugs

The gap it fills

- Existing analytics tools often need building-system integration
- Heat Check runs on plain meter exports
- Upload messy data: it maps columns, converts units, adapts to climate
- Fits campuses (or any organization with a building portfolio) without an energy team

Next steps

Roadmap

- Validate on real campus meter data
- Automatic weather lookup by location
- Verify savings after a fix is made

Try it

Live app:

heat-check-4w7d.onrender.com

Demo video:

youtube.com/watch?v=gHqYB2rRJCE

Code:

github.com/Data-Laur/Hackachusetts-Energy-Project

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